



TASK 3 – DBMS PROJECT

GROUP 7

GROUP MEMBERS:

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Neel Solanki

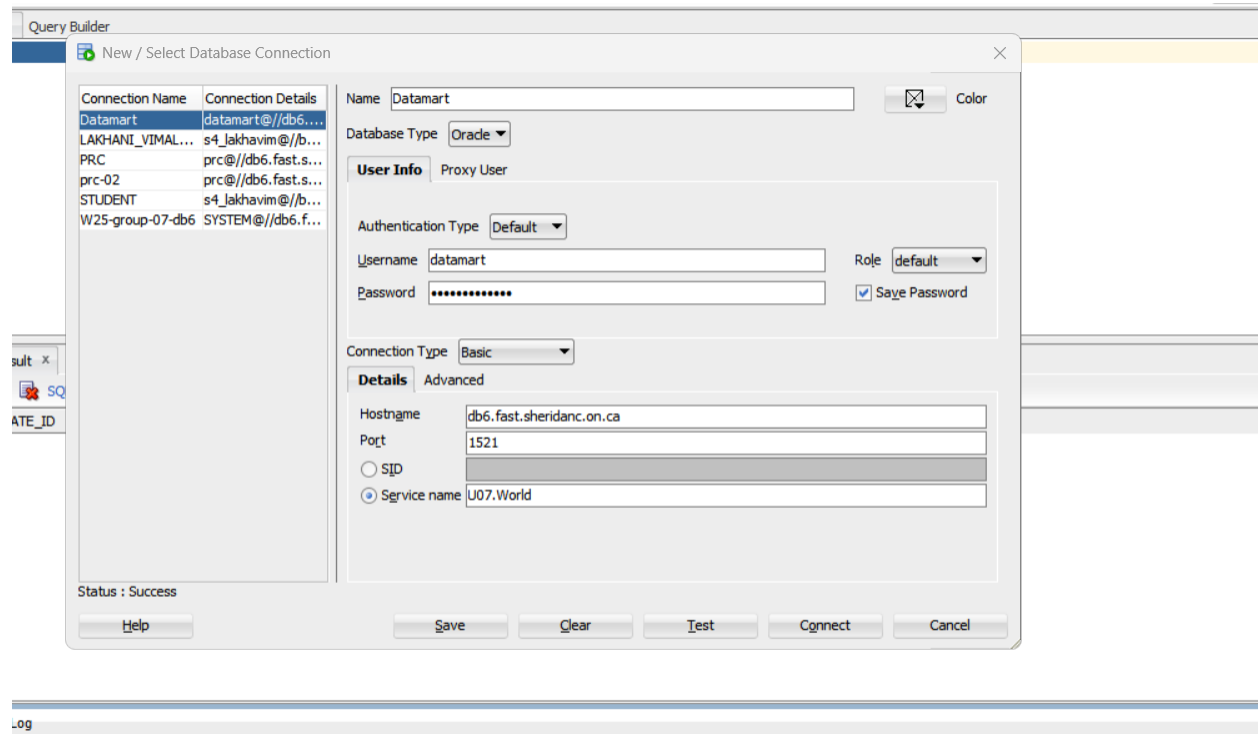
TASK 3.0.

TASK DISTRIBUTION

NEEL PARESH SOLANKI	Creates the connection to the Oracle DATAMART user and designs the FactDonations table, ensuring all foreign key relationships and schema grain are correctly implemented.
VIMAL SHANTIBHAI LAKHANI	Designs and creates all three dimension tables (DimDate, DimAddress, and DimVolunteer) with correct data types, surrogate keys, and appropriate hierarchy or descriptive fields.
PRATHAM MILANKUMAR DOSHI	Prepares and documents the ER diagram and schema design, including table relationships and grain explanation, and integrates this into the final project report.
RAKSHIT KHANNA	Validates schema integrity by testing joins between fact and dimension tables, inserts sample records, and supports ER diagram creation based on the implemented schema.

Creating a star schema in Oracle SQL Developer for Donation Table :

DATAMART CONNECTION:



CREATING REQUIRED TABLES:

1) **DimDate:**

CREATE TABLE DimDate (

date_id DATE PRIMARY KEY,

year NUMBER(4),

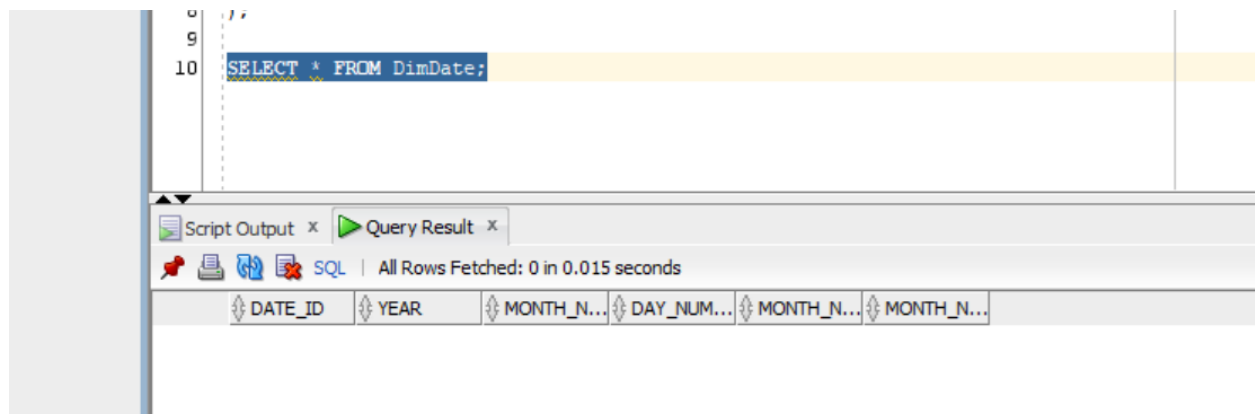
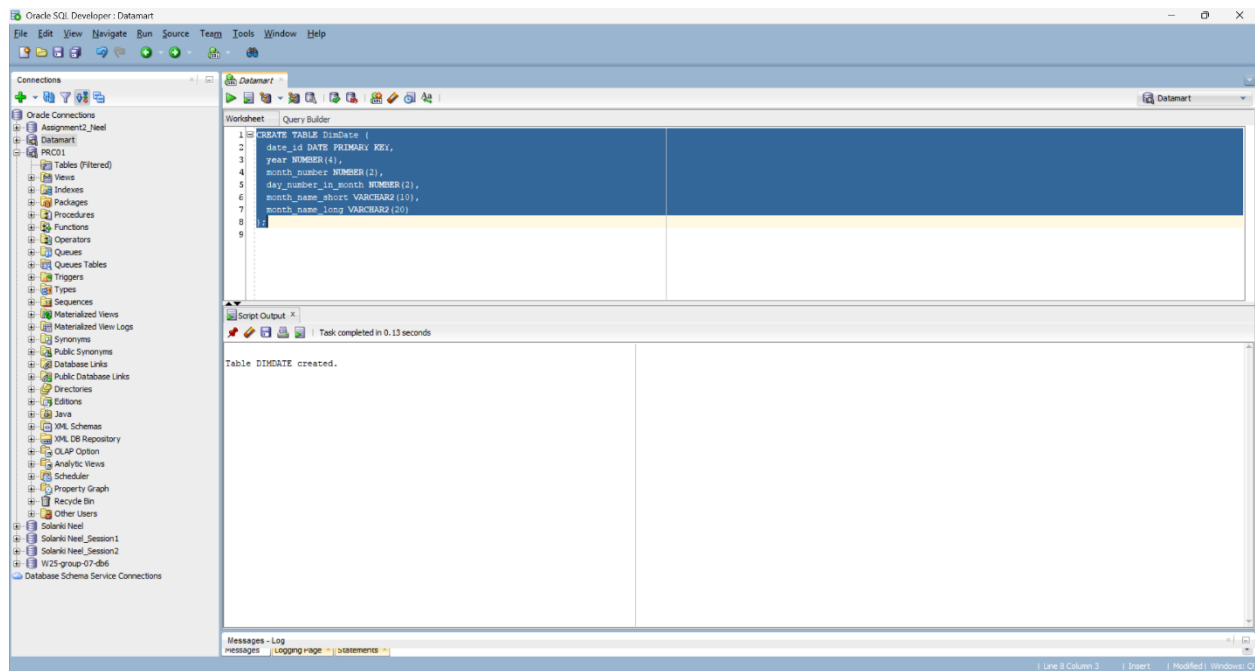
month_number NUMBER(2),

day_number_in_month NUMBER(2),

month_name_short VARCHAR2(10),

month_name_long VARCHAR2(20)

);

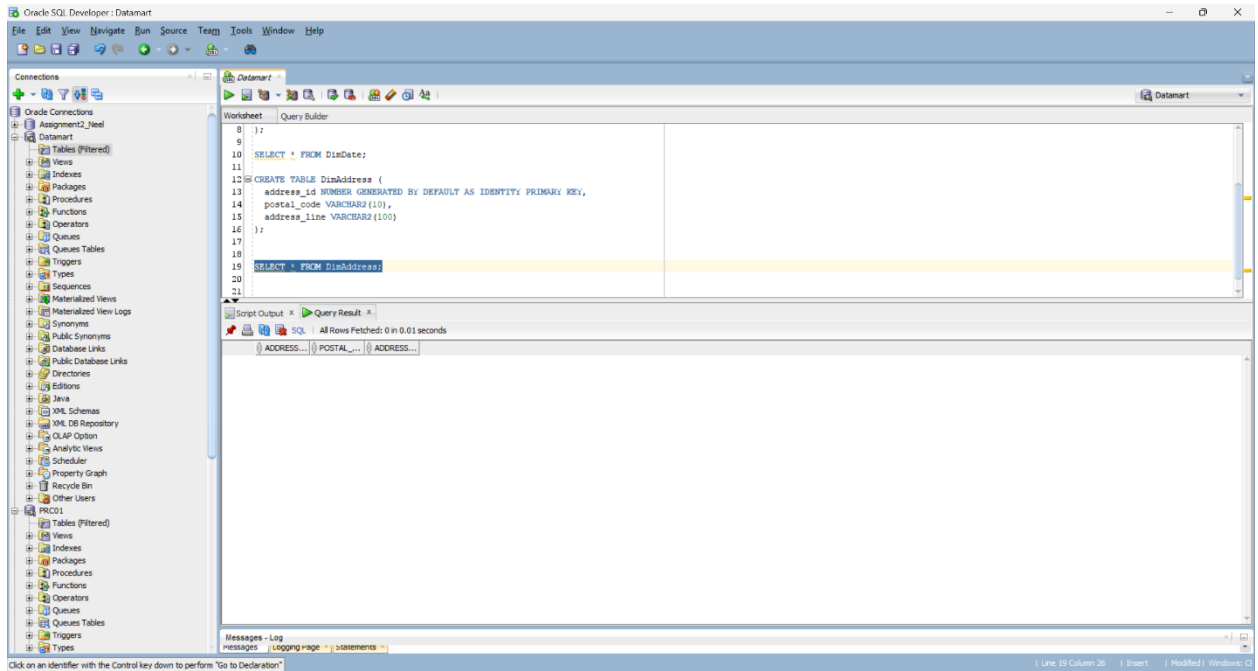
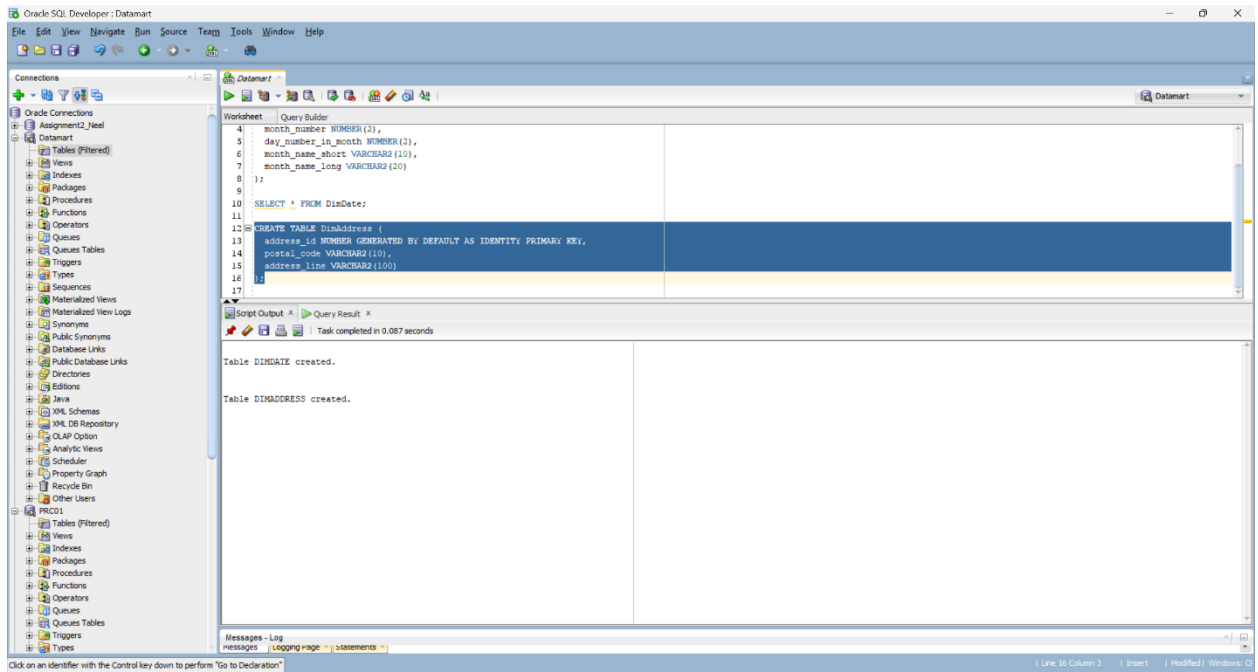


2) DimAddress :

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CREATE TABLE DimAddress (
  address_id NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
  postal_code VARCHAR2(10),
  address_line VARCHAR2(100)
);

```



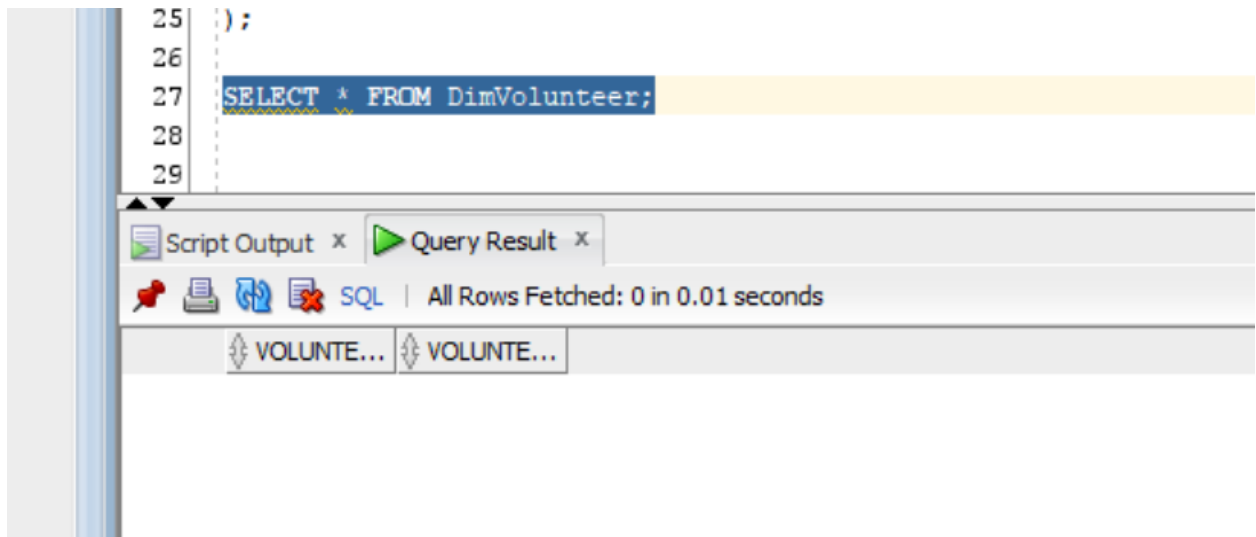
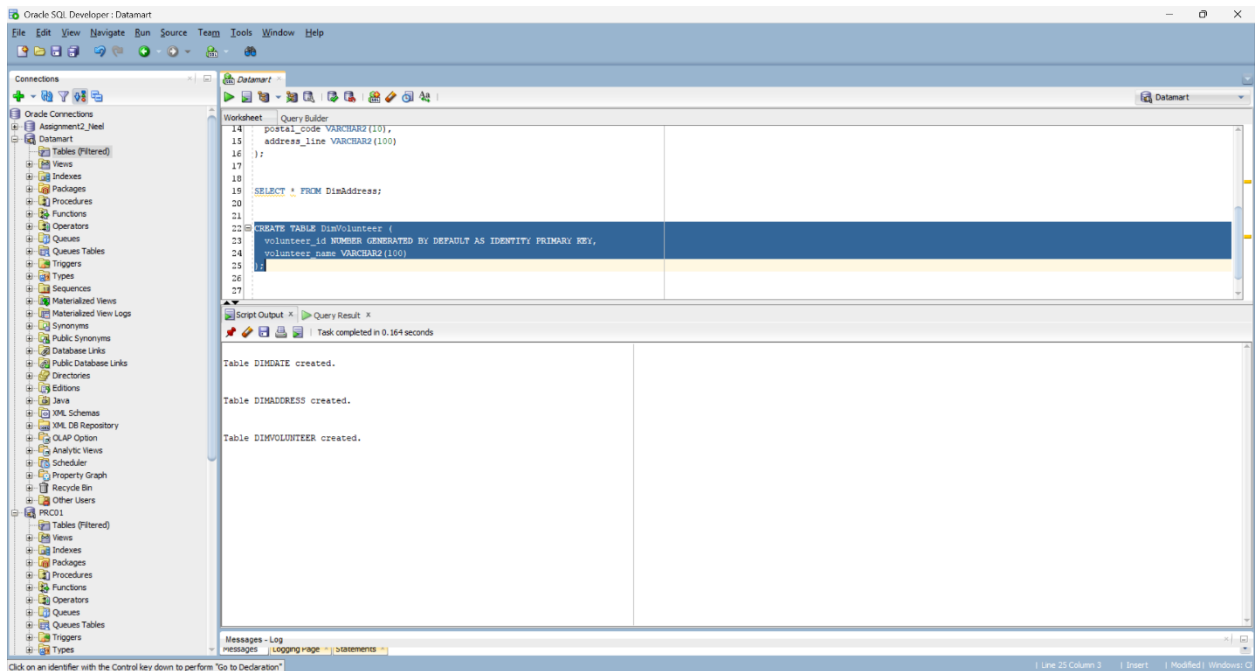
3) DimVolunteer :

CREATE TABLE DimVolunteer (

volunteer_id NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,

volunteer_name VARCHAR2(100)

);



4) FactDonations :

CREATE TABLE FactDonations (

date_id DATE,

address_id NUMBER,

volunteer_id NUMBER,

donation_count NUMBER,

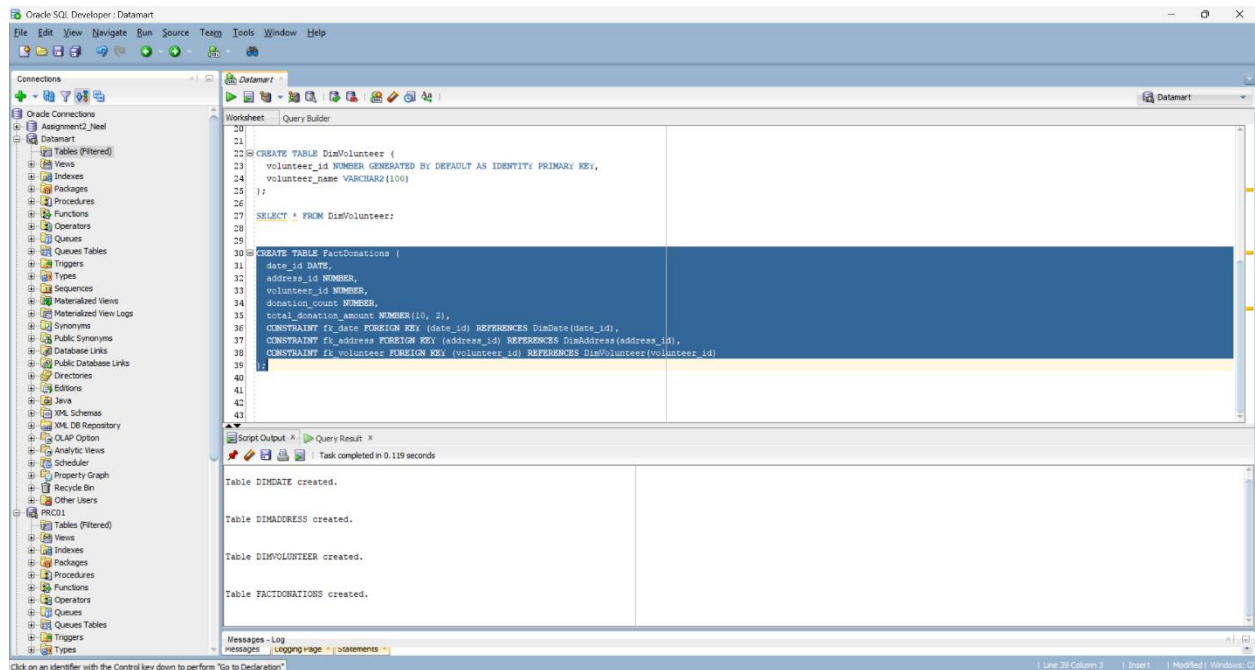
total_donation_amount NUMBER(10, 2),

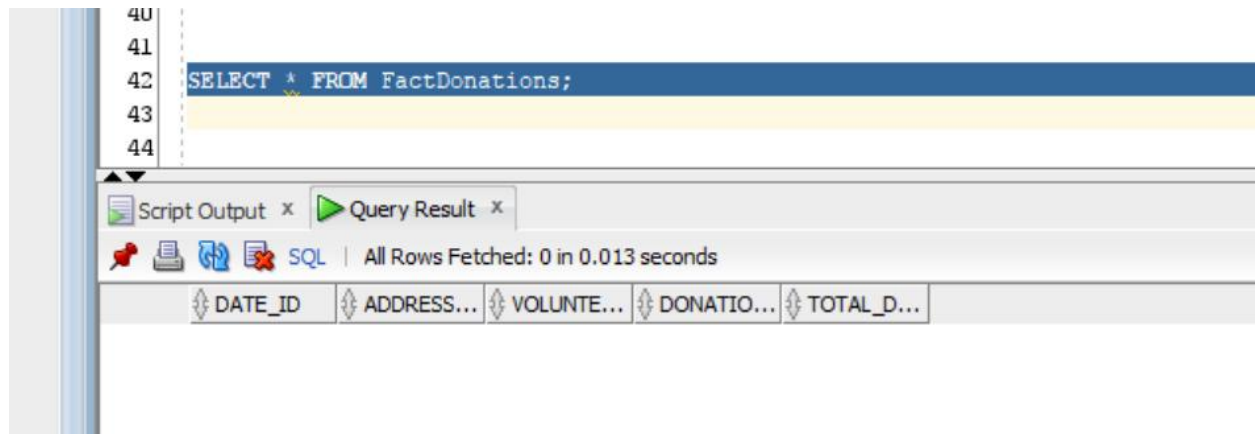
CONSTRAINT fk_date FOREIGN KEY (date_id) REFERENCES DimDate(date_id),

CONSTRAINT fk_address FOREIGN KEY (address_id) REFERENCES DimAddress(address_id),

CONSTRAINT fk_volunteer FOREIGN KEY (volunteer_id) REFERENCES DimVolunteer(volunteer_id)

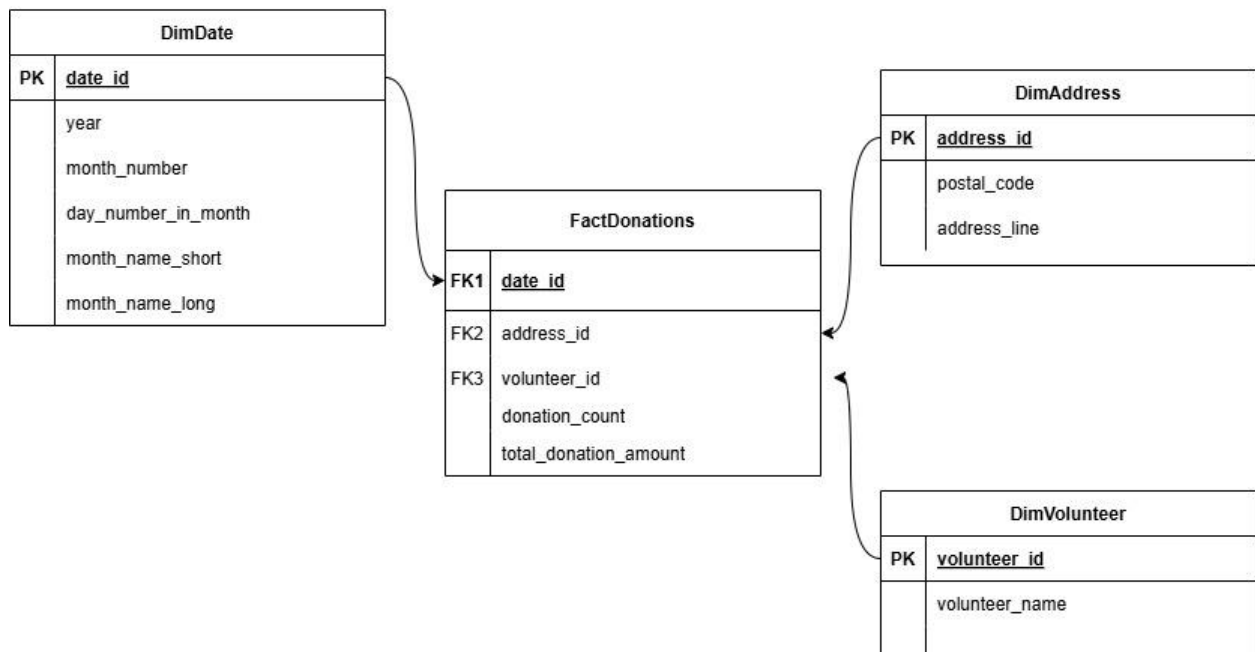
);





ER DIAGRAM:

Donations Data Mart – Star Schema ER Diagram



This Donations Data Mart uses a Star Schema with one central fact table and three dimension tables:

- **FactDonations:** Stores donation data (date, address, volunteer, count, amount).
- **DimDate:** Describes the donation date (year, month, day).

- **DimAddress: Describes the donation location (postal code, address).**
- **DimVolunteer: Contains volunteer details (name).**

Each dimension connects to the fact table via foreign keys for easy data analysis by time, location, and volunteer.